WIRELESS KEY SYSTEM FOR AUTOMOBILE VEHICLES

The project report submitted to

Veermata Jijabai Technological Institute, Mumbai

For the award of

DIPLOMA IN ELETRONICS ENGINEERING

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Academic year 2022-2023

DECLARATION

We certify that,

- The work contained in this report is original and has been done by me under the guidance of my guide.
- The work has not been submitted to any other Institute for the award of any diploma, or certificate.
- We have followed the guidelines of the Institute in preparing the thesis.
- Whenever we have used materials (data, theoretical analyses, figures, text, etc.) from other sources, we have given due credit to them by citing them in the text of the thesis and giving their details in the references Necessary.

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LIST OF ABBREVIATIONS

In alphanumeric order

SHORT FORM	LONG FORM
V	VOLTAGE
I	CURRENT
P	POWER
LCD	LIQUID CRYSTAL DISPLAY
DC	DIRECT CURRENT
AC	ALTERNATIVE CURRENT
RFID	RADIO FREQUENCY IDENTIFICATINS

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CHAPTER 1:- INTRODUCTION
1.1 INTRODUCTION

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The vehicle key is a safety device for the vehicle. Most keys on vehicles today still use physical keys, and some use keyless. This project examines the use of vehicle keys that are connected to smartphones. The method used in this research is literature study and experimental study. The technology used in this research is based on Arduino, and some of them are equipped with Bluetooth modules, HC-05, and relays.

This solution is an initial study of the SmartKey system concept. This SmartKey provides a function to turn on and turn off the machine. The resulting SmartKey from the results of this research will make it easier for application developers to control critical functions and be equipped with remote turning on and off methods, and develop various other features through application development.

The key generated in this study can be an alternative to lock system development in conventional motor vehicles.

"Automobiles can integrate additional wireless technologies like Bluetooth, Wi-Fi and Global Navigation Satellite System (GNSS) with advanced 3G/4G LTE technology."

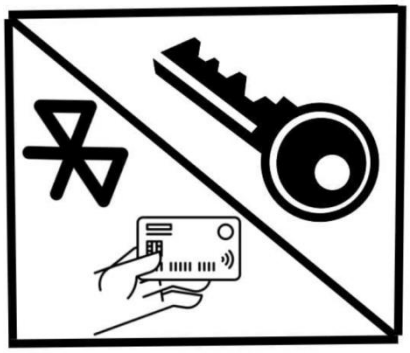

IMAGE ILLUSTRATING THE UNLOCKING METHODS OF AUTO VEHICLE
Fig 1.1

The future of the automotive industry lies in connecting vehicles with almost every physical thing in its surrounding and to the cloud. Wireless technologies are revolutionizing connectivity in the modern automobile. Ensuring that these technologies operate as expected has become increasingly important.

Wireless technologies for automotive advanced with wireless locking and unlocking of doors using apps, digital keys, etc. Later, the industry adopted secure and encrypted RF technology for lock/unlock mechanism of the automobile.

Bluetooth technology, which increased the convenience and comfort of the users. They could switch on/off appliances, like lights and horns, wirelessly. Later, it became a key feature for infotainment systems, apart from providing vehicle diagnostics to users through connected apps. The use of Bluetooth technology has improved the overall safety of auto vehicle users. Wireless connections relay high-quality audio, so that passengers can listen to their favorite music (stored on portable devices) and/or hook up their phones to the car.

Wi-Fi provides a robust data pipe for screen projection technologies. It enables a seamless transition for drivers, from their smartphones to cars of their own or rented from anywhere in the world.

Here we are operating the bike key system by unlocking in four ways:
1] physical key
2] RFID key
3] APP
4] digital key

Battery unit of auto vehicle
Fig 1.2

The temperature sensor can be used to see the overheating of the battery This applies in particular to the battery, as both icy cold and scorching heat can damage your battery and shorten its lifetime. This is due to the sensitive lithium-ion cells, which work best in terms of functionality and capacity within certain temperature windows. This is why manufacturers of e-bike batteries warn against letting them lie in the blazing sun or in a parked car that is heating up.
Instead of a temperature sensor, we have used a flame sensor for display purposes and overheating would cause the buzzer to beep.

Bike owners tend to focus on securing and monitoring the status of the bike remotely and planning their trips in accordance with the available stored on-board energy

1.2 EXISTING SYSTEM

Companies that specialize in wireless key systems specifically for bikes.
Some examples include:

- Velocate - A German company that offers a GPS bike tracking device with an integrated alarm system, controlled through a smartphone app.
- Linka - A US-based company that offers a smart bike lock that can be unlocked through a smartphone app or a Bluetooth key fob.
- Noke - A company that offers a Bluetooth-enabled smart bike lock that can be unlocked through a smartphone app.
- Lattis - A company that offers a smart bike lock with a built-in alarm, controlled through a smartphone app.
- BitLock - A company that offers a smart bike lock that can be unlocked through a smartphone app or a personalized code.

These companies and their products demonstrate how wireless key systems are being developed and implemented specifically for bike security, offering bike owners greater peace of mind and control over their bikes.
There are companies in India that offer wireless key systems for bikes. Some of the companies include:

- Steelbird: Steelbird offers a keyless ignition system for bikes called the Ignition Lock that uses wireless technology to start the engine.
- Resync: Resync offers a smart key system for bikes called the SmartKey, which allows riders to start and stop their bike without using a physical key, as well as control various features such as the lights and horn through a smartphone app.
- TVS: As mentioned earlier, TVS also offers a keyless ignition system called the SmartXonnect for its bikes, which allows riders to start and stop the bike without using a physical key.
- Hero MotoCorp: Hero MotoCorp offers a smart key system called the Hero Connect that allows riders to remotely start and stop their bike, as well as control various features such as the lights and horn through a smartphone app.

These are some of the companies in India that offer wireless key systems for bikes, and there may be others as well.

CHAPTER 2:- LITERATURE SURVEY

. 2.1 RESEARCH PAPER 1

ABSTRACT

Nowadays, Securing bikes from robbers and preventing unlicensed people from riding bike is an immense problem for all. Therefore, we established a sheltered solution for this problematic task. To avoid these tribulations, we proposed a ground breaking concept in our project. This system covenant with the design development of a theft control system for an automobile, which is being used to prevent the thievery of a vehicle. Our project makes use of an embedded system based on Global System for Mobile communication (GSM) technology. The designed & developed system is mounted in the vehicle. An interfacing GSM Module is also connected to the microcontroller (MSP430), which is connected to the engine. RF reader senses the License (RFID) information. This information is passed to the central processing system, RFID tag which contains the details of the driver, can be used as a security information and then the microcontroller unit sends SMS to the Vehicle owner. The microcontroller receives the signals from the RFID and it controls the ignition of the engine. Further, Alcoholic Sensor deducts whether the person has intoxicated, if the person is found inebriated, sensor deducts and makes microcontroller unit to limit the ignition system and the engine is locked. The designed system is very simple & cost is low.

INTRODUCTION:

These days vehicle thievery cases are higher than any other time; it has got to be fundamental to give a vehicle a sheltered security with the RFID and GSM. It is a vehicle security gadget that offers secured access to the vehicle. However, this framework could not demonstrate to give complete security to the vehicle in the event of burglary. So a more created framework makes utilization of an inserted framework focused around GSM innovation. A vehicle is stolen every six minutes in India. It is known that millions of people lose their vehicles due to theft and accidents. In addition to that there are many people lose their lives due to accidents on the roads. Most cases of theft have been caused by the lack of remote control system. It is known that, traditional systems used to monitor the vehicles, which depends mainly on alarm signal. Using GSM technology whereas this system also enhances personal safety of individuals. The main aim of the present research is to design and develop an advance and robust security system for vehicle that can prevent theft and provides information on robberies. The system being developed through the present work uses GSM system and can be made affordable. Major bike accidents occur due to unlicensed and drunken people who misuse driving laws. Therefore, To overcome this we found a secure solution to this problem. We provide bike access only to those people who own a license provided with RFID tag and who has not drunk. We also help the vehicle owner to get the details of the person who gets access to bike, by giving a text message to the registered mobile number through GSM technology. The output is also obtained from the LCD display, which is interfaced to the embedded kit.

MATERIAL:

Hardware Required

- MSP430
- GSM Module
- RFID Reader and RFID Tag
- Relay
- DC Motor (Vehicle)
- Ignition Key
- Sensors

Power Supply MSP430 MICROCONTROLLER: The definition of a microcontroller is somewhat difficult due to the constantly changing nature of the silicon industry. What we today consider a microcontroller with medium capabilities is several orders of magnitude more powerful, than the computer used on the first space missions. Nevertheless, some generalizations can be made as to what characterizes a microcontroller. Typically, microcontrollers are selected for embedded systems projects, i.e., control systems with a limited number of inputs and outputs where the controller is embedded into the system.

RFID READER: Radio frequency identification (RFID) is a general term that is used to describe a system that transmits the identity (in the form of a unique serial number) of an object wirelessly, using radio waves. RFID technologies are grouped under the more generic Automatic Identification (Auto ID) technologies. The barcode labels that triggered a revolution in identification systems long time ago, are inadequate in an increasing number of cases. They are cheap but the stumbling block is their low storage capacity and the fact that they cannot be reprogrammed.

RFID READER & TAG

RFID TAG: An RFID tag is composed of an antenna, a wireless transducer, and an encapsulating material. These tags can be either active or passive. While the active tags have on-chip power, passive tags use the power induced by the magnetic field of the RFID reader. Thus passive tags are cheaper but with a lower range (<10mts) and more sensitive to regulatory and environmental constraints, as compared to active tags. GSM: GSM (Global System for Mobile Communications: originally from

Group Special Mobile

is the most popular standard for mobile phones in the world. Its promoter, the GSM Association, estimates that 80% of the global mobile market uses the standard GSM is used by over 3 billion people across more than 212 countries and territories. Its ubiquity makes international roaming very common among mobile phone operators, enabling subscribers to use their phones in many parts of the world. GSM differs from its predecessors in that both signaling and speech channels are digital, and thus is considered a second-generation (2G)

mobile phone system. This has also meant that data communication was easy to build into the system.

GSM

BLOCK DIAGRAM

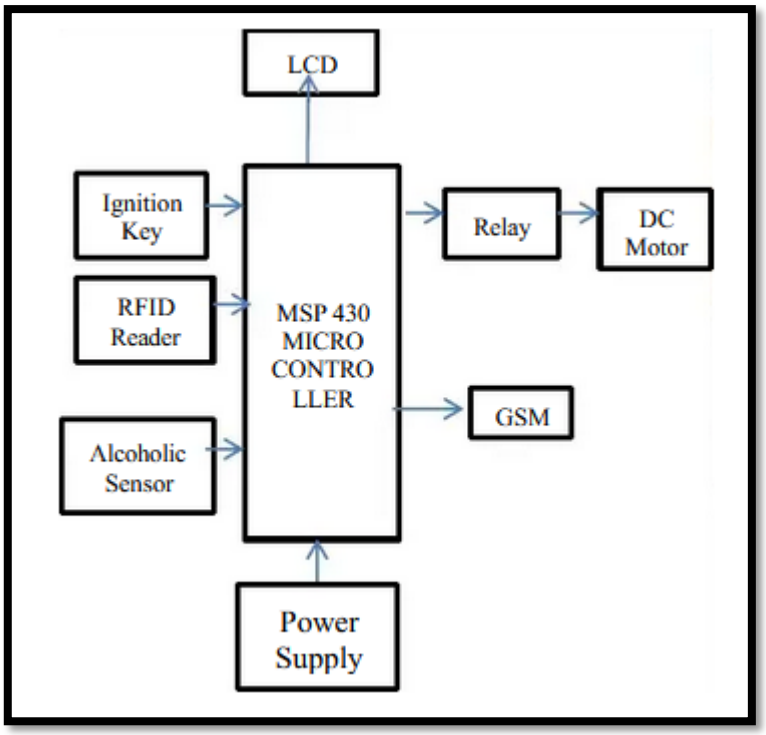

Fig 2.1

OBJECTIVES

The Objective is to design and develop an advanced vehicle security and locking system in the real time environment. This system is made for security using RFID and also we can lock the vehicle.

To design and develop a simple and low cost bike security system.

CONCLUSION:

We have implemented a simple system using RFID & GSM.
It is a low cost & design efficient system.
Low in power consumption.
The secured & limited access to bikes is implemented.

The information about the persons who are accessing the bike is got as SMS through GSM Technology.

2.2 RESEARCH PAPER 2

TWO WHEELER VEHICLE SECURITY SYSTEM

RELATED WORK

Indeed, we are not the first to observe the flaws and limitations of the present-day vehicle security systems. Several researchers have described potential vulnerabilities in automotive security systems. The traditional security systems as priced low, but they merely act as an alarm system and are no match to the well-equipped thief. Many security systems have been proposed over the years, but almost all the recent advanced security systems are designed especially for cars. Several researchers have even used image processing technology to capture the face of driver.

and compare it with the picture of authorized drivers to detect the intrusion some proposed systems include a fingerprint detection system [19] along with face detection. These security systems are complex, costly and cannot be implemented on two-wheelers. Two-wheeler vehicles offer very less space to install the security module and hence even area is one of the major constraints. The demand is to design a system that performs the necessary functions, and small enough to be placed under the seat of the vehicle. simple to operate, reasonably priced and small enough to be placed under the seat of the vehicle.

III. DESCRIPTION OF PROPOSED TWO WHEELER VEHICLE SECURITY SYSTEM

An overview of the complete system is described in this section before detailing the specifications and the necessity for each module in the system. The general view of operation of the proposed security system.

The conventional handle locking system is replaced by a handle lock operated by a servo motor and controlled by the Remote Keyless System (RKS). RKS has a transmitter and a receiver module. The receiver module is installed on the vehicle and the owner of the vehicle has the remote (transmitter module). The RKS remote could be used to lock/unlock the locking system (handle lock, fuel lock and rear wheel lock), switch ON/OFF the engine and to turn off the alarm. The fuel lock replaces the knob at the nozzle of fuel tank. The real wheel lock is used to jam the sprocket of the wheel. GSM Tower Satellite Vehicle Owner’s Mobile Remote

The vehicle owner’s cell phone with the registered Subscriber Identity Module (SIM) number acts as the master key of the security system. It is given higher priority over RKS and hence it can override the instructions from RKS. A Short Message Service (SMS) is sent to the registered SIM via the Global System for Mobile Communication (GSM) module whenever the vehicle is unlocked using the RKS remote. The owner can in turn send a SMS to initiate the locking sequnce, if he/she feels an unauthorized person has unlocked the vehicle [13]. Global Positioning System (GPS) module is used to track the location and monitor the speed of the vehicle. Piezoelectric sensors are used as vibration sensors to detect any tampering with the vehicle. Two sensitivity levels are chosen so as to nullify the chance of false alerting. The owner is alerted by an SMS and a loud alarm is activated if there is

any physical tampering with the vehicle or if the vehicle has been moved from the place where it was parked (>15m). The owner can, anytime, request the security system of the vehicle for its status by sending an SMS. The status message sent in response to the request includes the lock status, location coordinates and the speed of the vehicle.

Infra-Red (IR) sensors are used as side stand indicators, until and unless the side stand is not pulled up the engine cannot be started.

HARDWARE MODULE

Remote Keyless System (RKS)

RKS module is designed to operate at a frequency of 433 MHz with a half-duplex communication link. The Remote consists of a 433 MHz transmitter, encoder, power supply, and a simple loop antenna. The receiver module consists of a 433 MHz receiver, decoder, power supply, and a simple loop antenna. the receiver module is interfaced with the MCU. The remote consists of four buttons, each associated with a different operation. The chosen pair of encoder and decoder should have the same number of addresses and data format. We use a 4-bit hot code to represent each button. The 12-bit parallel data is encoded to serial data for transmission through RF transmitter. These 12 bits are divided into 8 address bits and 4 hot code bits. This 8-bit address data is unique to every pair of receiver and transmitter modules as stated earlier. The received serial data is decoded into parallel data. The 8-bit address data is compared with the local address data three times continuously. If these two address bits match, the 4-bit hot code is made available to the MCU. For our design (under standard conditions), we obtained a maximum communication range of approx. 100m, justifying the use of a low-cost simple loop antenna.

Locking system

The handle, fuel and rear wheel locking systems are operated by servo motors. The Servo motor is chosen because the output shaft can be rotated to a specific angular position by sending it a pulse width modulation (PWM) signal through the control wire. Atmega328p-pu MCU has 6 inbuilt PWM which are used to generate the PWM signal to control the servo motor. The operational voltage of 6V and current of 1A is supplied from the driver IC (L293D) as shown in Fig. 2. While parking the vehicle, the rider should align the handle to the extreme left and press the lock button on the remote. This initiates the locking sequence of the handle, fuel and rear wheel lock. To avoid fuel theft, a lock is placed at the nozzle of the fuel tank. This lock acts as a valve. Once the vehicle is locked, the fuel lock blocks the nozzle, disrupting the fuel flow in the tube. As there is no fuel flowing into the engine, it cannot be started, thus serving two purposes. Another lock is placed at the rear side of the vehicle, to jam the sprocket of the wheel. Once the sprocket is jammed, the wheel cannot rotate. This ensures redundancy in the proposed security system. The owner can control the locking system from his/her mobile phone (sending an SMS) or RKS remote.

GPS Module

Gms-u1LP GPS module is used for tracking the vehicle. It utilizes MediaTek GPS MT3329 that supports up to 66 channels of satellite searching with -165dBm sensitivity and 10Hz maximum update rate for precise GPS signal processing under low receptive and high-velocity conditions. It has a power-saving switching mode power supply (SMPS) that helps reduce the overall GPS power consumption. We used the National Marine Electronic Association (NMEA) 0183 protocol which is a worldwide accepted standard. The Recommended Minimum Navigation Information GPRMC sentence is decoded to get the coordinates, speed, UTC time, date month, year, and other necessary information.

GSMModule

The GSM module is required to establish a communication link between the owner of the vehicle and the security system. We used the SIM300 GSM module in our system. AT commands were used to control this module. SIM300 is a Tri-band GSM/GPRS engine that works on frequencies EGSM 900 MHz, DCS 1800 MHz, and PCS1900 MHz. SIM300 also provides General Packet Radio Service (GPRS). The current consumption is as low as 2.5mA in SLEEP mode. SIM memory is used to store messages. The SIM300 module communicates with the MCU using asynchronous serial communication with a baud rate of 9600. The owner of the vehicle can send an SMS to lock down the bike. The owner can also disable the remote keyless system if the remote is stolen.

These features become very useful if in case duplicate or stolen remote is used to unlock the vehicle. Thus the cell phone of the owner with the registered SIM acts as the master key, which can override the instructions from the remote.

Engine Immobilizer

This module is used to control the ignition system of the vehicle, thereby controlling the engine. The capacitive discharge ignition (CDI) module uses capacitor discharge current output to fire the spark plugs. The capacitive discharge ignition (CDI) module has 6 pins in common. Of these 6 pins, one pin is for the ignition trigger pulse, one pin is to be connected to the 12V ignition coil, one pin is for the supply of ignition power to spark plugs, one pin is to be connected to the kill switch and the remaining two pins are to be grounded. If the kill pin is grounded then the ignition power pin does not supply 20,000V to the spark plug, which is essential to produce a spark in the power stroke. A relay is used to accomplish this function. Relay is used in “normally open mode”, one terminal is connected to the kill pin and another is grounded. Whenever the engine has to be immobilized the microcontroller unit (MCU) activates the relay coil which shorts the kill pin to the ground.

Vibration sensor

A piezoelectric sensor is used for this application, as the cost of this sensor is very less and it has reasonably good sensitivity. The chassis runs throughout the vehicle and it is made of metal, so the vibration in any part resonates all over it. Piezoelectric sensors are placed on the chassis, one at the front end and another at the at rear end. As and when there is a vibration (because of the intruder tampering with the vehicle, it is transferred to the piezoelectric film and there is a proportional voltage developed at the terminals of the piezoelectric sensor. The analog voltages are converted to digital values by the inbuilt 10-bit Analog to Digital Converter (ADC) of the Atmega328p-pu MCU. The voltage produced is directly proportional to the intensity of the vibration. A threshold value is set to distinguish between the high and low intensities

Software module

The Arduino integrated development environment (IDE) v1.52 was used for writing the code. The Arduino programming language is an implementation of Wiring, a similar physical computing platform, which is based on the Processing multimedia programming environment. Arduino is an open-source electronics prototyping platform based on flexible, easy-to-use hardware and software. Arduino Uno R3 board was used as a programmer to dump the boot loader and the code into the MCU (Atmega328p-pu). In this section, the flowchart for individual modules is described. These are basically the subroutines that are called from the main program whenever a particular task is to be performed. Once the subroutine is executed, the control returns to the main program.

GSM subroutine

The GSM subroutine is called only when the GSM hardware receives a SMS from the registered SIM number. every SMS is associated with a particular set of operations. If the status request is made by

owner then the status of locking system, location and speed is sent in response. If Lock SMS is sent by owner then the lock sequence is initiated and RKS remote is disabled. If Unlock SMS is sent by owner then the unlock sequence is initiated. If the Enable SMS is sent then the remote is reactivated. If the Buzzer SMS is sent then the buzzer is deactivated. Thus the GSM module enables the owner to control the security system of the vehicle.

CONCLUSIONS

Our proposed Two Wheeler Vehicle Security System is the advanced, reliable, and robust version of the security mechanism for two-wheeler vehicles. The proposed security system also gives space, in terms of hardware and software, to add up custom applications to make the product even more user-friendly. The proposed TWVSS can be installed on two-wheeler vehicles of any class or company, thereby creating a huge market for the product. Stress was laid in designing a cost-efficient system so that it could also be even bought by the owners of low-end bikes. The small size of the module allows it to be placed under

CHAPTER 3:- MARKET SURVEY

In the automotive smart key market, the growth of the Asia-Pacific region is expected to be more during the forecast period. China is one of the biggest automotive markets in the world. In 2018, the volumes of production and sales of vehicles were lower than expected at the beginning of the year, due to political factors and microeconomics. However, new energy vehicle (NEV) sales continued to rise rapidly, while exports also showed fast growth.

The demand for smart keys in the Chinese, as well as the Indian market, is growing rapidly, as various Chinese and Indian companies are launching vehicles with keyless entry features, which shall lead to a growth of the automotive smart key market in China and India. The increasing demand for smart keys with advanced technology in Japan is likely to propel the smart key market in the country in the years to come. Thus, such developments are expected to lead to growth in the automotive smart key market in the Asia-Pacific region.

Competitive Landscape

The automotive smart key market is fragmented, with various players existing in the market. Tokai Rika Co. Ltd is the leading player in the automotive smart key market. Additionally, Tokai Rika, Denso, and Alpha Corporation have been the major suppliers of electronic keys/smart keys/keyless systems to Toyota, Nissan, etc., in Japan. Other players in the market include Bosch, Valeo, etc., which constitute over 20% of the market.

In addition, the majority of these players are now offering smart key solutions in the aftermarket channel. Various initiatives, as well as product innovations done by Tokai Rika Co. Ltd have led them to strengthen their presence in the market. For example, in 2017, the company Continental AG's Keys-as-a-Service Portfolio led to car owners possessing the ability to access cloud-based solutions, like Continental Smart Access (CoSmA) and Remote Cloud Key (RCK). This solution shall help owners to secure keyless vehicle access, as well as authorization, just by making use of a smart device, such as a smartphone.

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CHAPTER 4:- DESIGN OF EXPERIMENTAL SET UP

4.1: BLOCK DIAGRAM

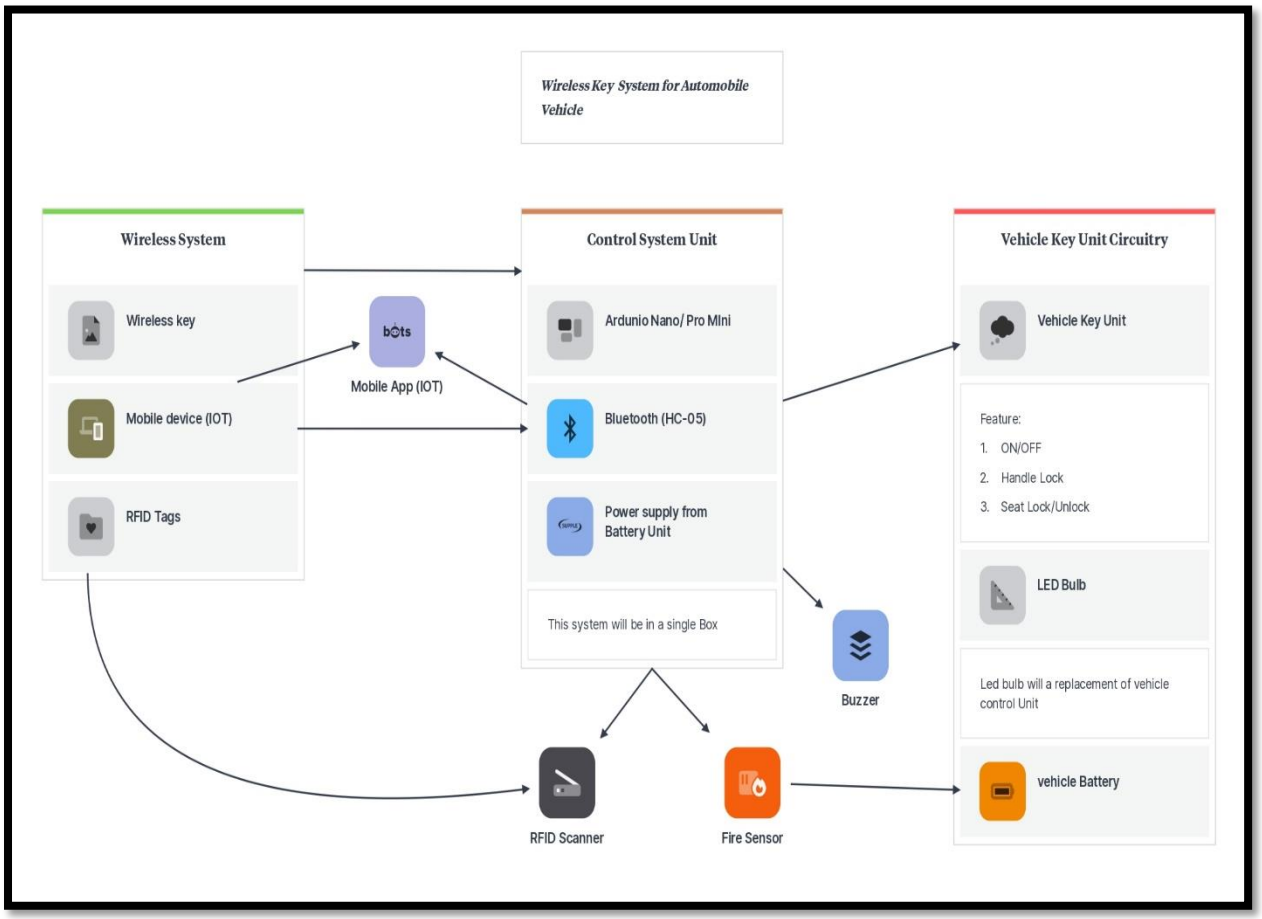

Fig 4.1

4.2 SCHEMATIC DIAGRAM

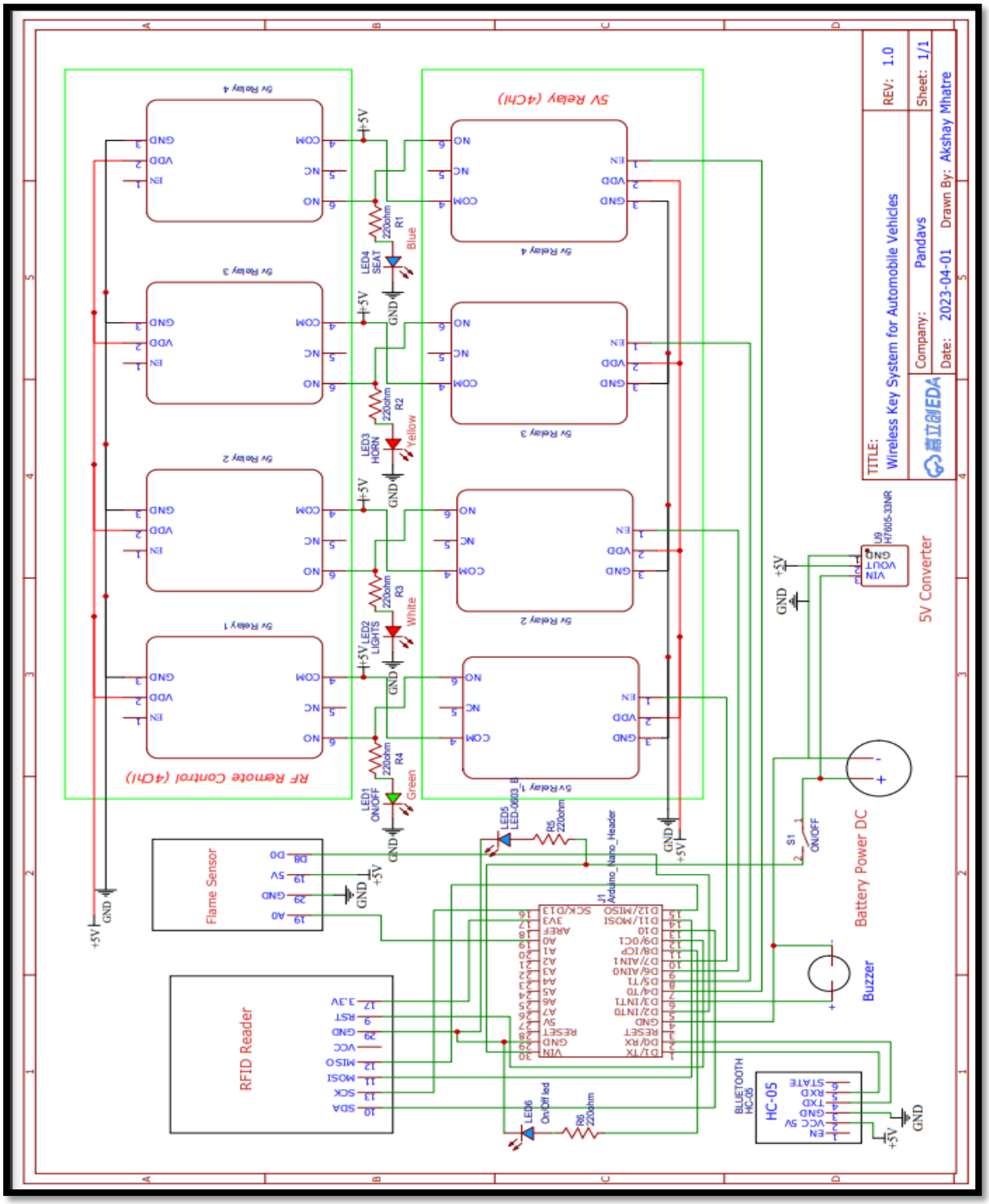

Fig 4.2

CHAPTER 5:- DEVELOPMENT OF EXPERIMENTAL SETUP

5.1] APP DEVELOPMENT

The app was developed using the MIT app inventor.
A brief intro to MIT app inventor:

Fig 5.1

In MIT App Inventor, users code application behavior using a block-based programming language. There are two types of blocks in App Inventor: built-in blocks and component blocks. The built-in blocks library provides the basic atoms and operations generally available in other programming languages, such as Booleans, strings, numbers, lists, mathematical operators, comparison operators, and control flow operators.

Developers use component blocks (properties, methods, and events) to respond to system and user events, interact with the device hardware, and adjust the visual and behavioral aspects of components.

How to use it??

1) Connect the auto vehicle to the mobile app using Bluetooth.
2) After the connection has been made the user can start the vehicle and also have control of the seat, lights, and buzzer.
3) This is simply done by pressing on-off buttons.
4) This app can be simply downloaded by scanning the QR code or also we can use the MIT companion.

It is a simple app consisting of two screens:

1) Splash screen
2) Bluetooth connection and controlling

App development procedure:

1) Making of the splash screen:

Fig 5.2

A splash screen, also known as a launch screen, is the first screen that a user sees when opening your app, and it stays visible while the app is loading. Splash screens only appear briefly on your screen; if you turn your head away, you can miss them. Typically, you'll notice the app name and its version with fast slider.

The app logo was downloaded using the flaticon website and also the background were chosen in contrast with the app logo. These background were simply pasted from google.

Adjustments were done in the images , horizontal arrangements, labels ,textbox,etc to bring them in a correct alignment to make and appear the splash screen in a decent way.

How it was made:

1} Click on screen1 and make the vertical and horizontal arrangement at the center.

2} Drag a image block at the screen and add the logo with correct height and width arrangements. Add a label to highlight the name of the app and its version.

3} Drag a slider to show progressing bar and add a non visible component clock.

4} Go to the block section to add to the functionality of splash screen.

Blocks Editor:

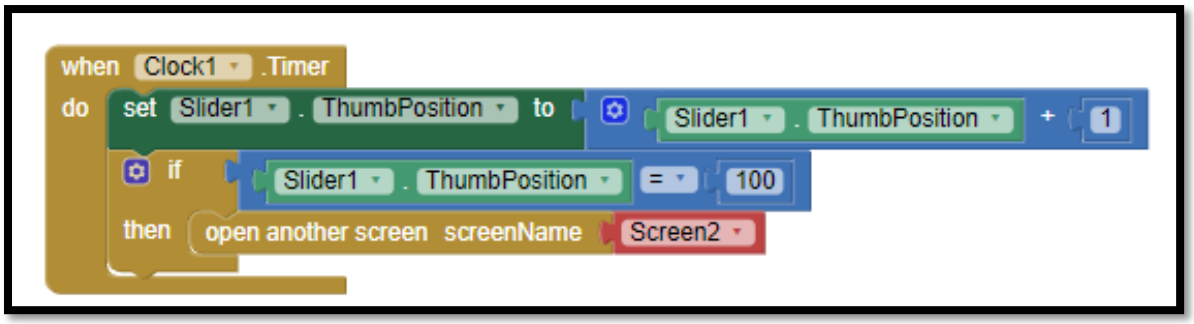

Fig 5.3

2) Proceeding to the connection and control screen:

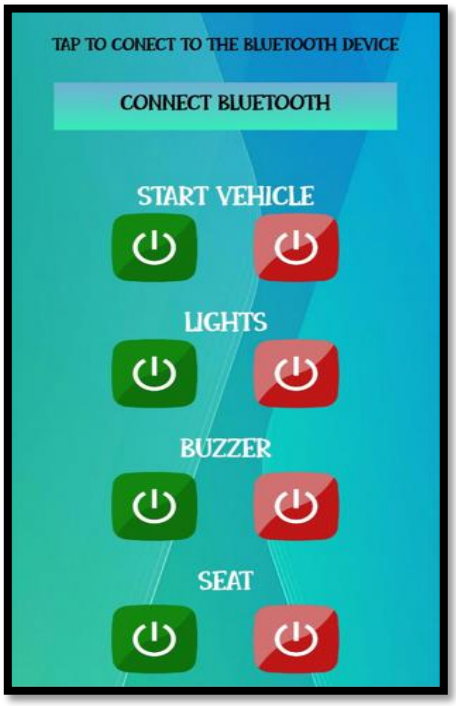

Fig 5.4

Bluetooth established a very low power, short range communication link between two devices. The app framework provides access to the Bluetooth functionality through Bluetooth APIs. These APIs let apps connect to other Bluetooth devices, enabling point-to-point and multipoint wireless features such start of vehicle, control on other functions such as seat,lights and buzzer.
MAKING:

1) Select the ListPicker component from user interface. ListPicker is the button, which displays the list of available bluetooth devices and handle the selection.
2) Click on the Layout from palette and select Horizontal arrangement. Horizontal arrangement component is to display a group of components laid out from left to right. We are using this component to display a button horizontally which is “Connect”.
3) Rename the buttons from “Button1 to Connect”
4) At last, connectivity from palette select Bluetooth Client and drag and drop to the viewer panel. Note that, this is the hidden component.
5) After connection has been made display it as ‘connected’.

6) The using the texts and labels add the four functions with their on off functionality. Functions are as follows: start vehicle,seat,lights and buzzer
7) Go to the blck editor section.

Block editor:
For Bluetooth connection:

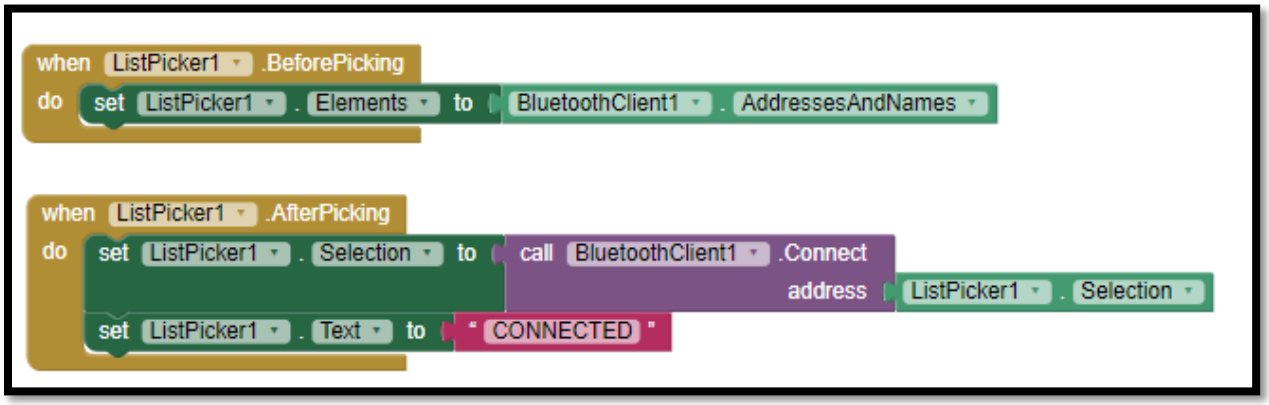

Fig 5.5

For control:

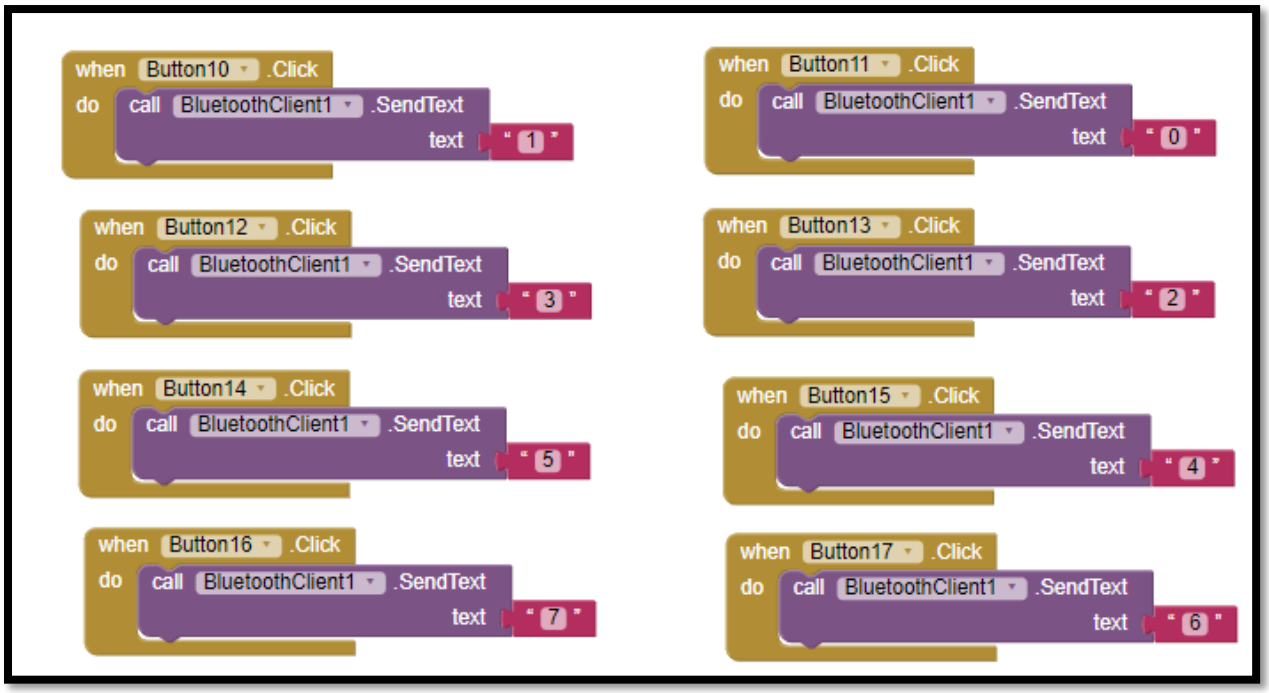

Fig 5.5

NOTE: The arduino program should be configured as per the data entered in the text.
This app was tested by making a small Bluetooth circuit using HC-05 on breadboard and connecting leds to check out their proper working.

5.3] CODE

```
#include "SPI.h"
#include "MFRC522.h"

#define SS_PIN 10
#define RST_PIN 9

MFRC522 rfid(SS_PIN, RST_PIN);
MFRC522::MIFARE_Key key;

//Flame Sensor//
const int flamepin=A0;
const int buzpin=3;
const int threshold=500;// sets threshold value for flame sensor
int flamesensvalue=0;

//BlueetoothApp//
const int LED1 = 4; // Pin for LED1
const int LED2 = 5; // Pin for LED2
const int LED3 = 6; // Pin for LED3
const int LED4 = 7; // Pin for LED4

//RFID//
int ONOFFled = 8;
int j=0;

String RFIDKey1 = "D3:83:9D:00"; //assign accepted RFID Tag 1
String RFIDKey2 = "95:1E:C8:73"; //assign accepted RFID Tag 2

void setup() {
  Serial.begin(9600);
  SPI.begin();
  rfid.PCD_Init();
  pinMode(ONOFFled, OUTPUT);
  pinMode(flamepin,INPUT);
  pinMode(buzpin,OUTPUT);
  pinMode(LED1, OUTPUT); // Set LED1 as an output
  pinMode(LED2, OUTPUT); // Set LED2 as an output
  pinMode(LED3, OUTPUT); // Set LED3 as an output
  pinMode(LED4, OUTPUT); // Set LED4 as an output

}

void loop()
{
  {
  if (Serial.available() > 0) { // Check if data is available
    char data = Serial.read(); // Read the incoming data

    if (data == '1') { // If data is '1', turn on the LEDs
      digitalWrite(LED1, HIGH);
```

```
    }
    else if (data == '0') { // If data is '0', turn off the LEDs
      digitalWrite(LED1, LOW);
    }

  if (data == '3') { // If data is '1', turn on the LEDs
      digitalWrite(LED2, HIGH);
    }
    else if (data == '2') { // If data is '0', turn off the LEDs
      digitalWrite(LED2, LOW);
    }

  if (data == '5') { // If data is '1', turn on the LEDs
      digitalWrite(LED3, HIGH);
    }
    else if (data == '4') { // If data is '0', turn off the LEDs
      digitalWrite(LED3, LOW);
    }

  if (data == '7') { // If data is '1', turn on the LEDs
      digitalWrite(LED4, HIGH);

    } else if (data == '6') { // If data is '0', turn off the LED
      digitalWrite(LED4, LOW);
    }
  }
}
flamesensvalue=analogRead(flamepin); // reads analog data from flame sensor
Serial.println(flamesensvalue);
if (flamesensvalue<=threshold) { // compares reading from flame sensor with the threshold
value
tone(buzpin,100);
delay(100); //stops program for 1 second
}
else{
noTone(buzpin);
}
{
  if (!rfid.PICC_IsNewCardPresent() || !rfid.PICC_ReadCardSerial())
    return;

  // Serial.print(F("PICC type: "));
  MFRC522::PICC_Type piccType = rfid.PICC_GetType(rfid.uid.sak);
  // Serial.println(rfid.PICC_GetTypeName(piccType));

  // Check is the PICC of Classic MIFARE type
  if (piccType != MFRC522::PICC_TYPE_MIFARE_MINI &&
    piccType != MFRC522::PICC_TYPE_MIFARE_1K &&
    piccType != MFRC522::PICC_TYPE_MIFARE_4K) {
    Serial.println(F("Your tag is not of type MIFARE Classic."));
```

```
  return;
 }

 String strID = "";
 for (byte i = 0; i < 4; i++) {
  strID +=
  (rfid.uid.uidByte[i] < 0x10 ? "0" : "") +
  String(rfid.uid.uidByte[i], HEX) +
  (i!=3 ? ":" : "");
 }
 strID.toUpperCase();

 Serial.print("Tap card key: ");
 Serial.println(strID);

 //check RFID keys
 if(strID.indexOf(RFIDKey1)>=0 or strID.indexOf(RFIDKey2)>=0){

  if(j==2){j=0;}
  if(j==0){
   digitalWrite(ONOFFled, HIGH);
   digitalWrite(LED1, HIGH);
   }

  if(j==1){

   digitalWrite(ONOFFled, LOW);
   digitalWrite(LED1, LOW);
   }
   j++;
  }
  else{

   digitalWrite(ONOFFled, LOW);
   digitalWrite(LED1, LOW);
  }

 rfid.PICC_HaltA();
 rfid.PCD_StopCrypto1();
}
}
```

CHAPTER 6:- APPARATUS USED IN EXPERIMENTAL WORK

6.1 LIST OF COMPONENTS

SR no.	Components	Specifications	Quantity	Price
1.	Bike Key unit	-	1	-
2.	Arduino Nano	-	1	-
3.	RFID Card And Reader	-	1	-
4.	Bluetooth module	HC-05	1	-
5.	Relay	5V, 4 - Channel	1	-
6.	General Purpose PCB	-	1	-
7.	Flame sensor	-	1	-
8.	Battery Unit	-	1	-
9.	Buzzer	-	1	-
10.	RF Remote Control	-	1	-
11.	On/Off switch	(pushbutton switch)	1	-
12.	Headers	(Male&Female)	As per need	-
13.	Connenctors		As per need	-
14.	Plastic box	(8*8)	1	-
15.	Led’s	White,Red,Yellow,Green	As per need	-
16.	Wires	(24gauge)	As per need	-
17.	Female connector (key Unit)	-	1	-

6.2 HARDWARE REQUIREMENTS

- Rfid

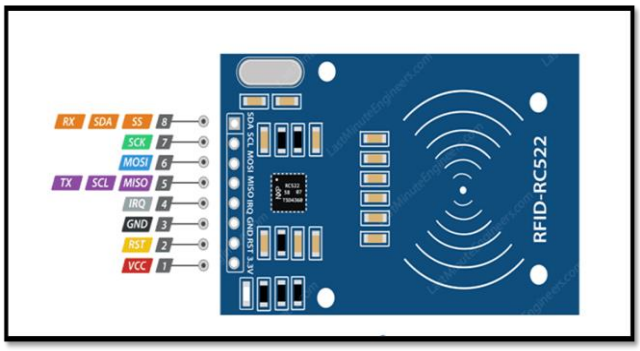

Fig 6.1

An Rfid Or Radio Frequency Identification System Consists Of Two Main Components, A Tag Attached To The Object To Be Identified, And A Reader That Reads The Tag.
A Reader Consists Of A Radio Frequency Module And An Antenna That Generates A High Frequency Electromagnetic Field. Whereas The Tag Is Usually A Passive Device (It Does Not Have A Battery). It Consists Of A Microchip That Stores And Processes Information, And An Antenna For Receiving And Transmitting A Signal.
When The Tag Is Brought Close To The Reader, The Reader Generates An Electromagnetic Field. This Causes Electrons To Move Through The Tag's Antenna And Subsequently Powers The Chip. The Chip Then Responds By Sending Its Stored Information Back To The Reader In The Form Of Another Radio Signal.

- Arduino Nano

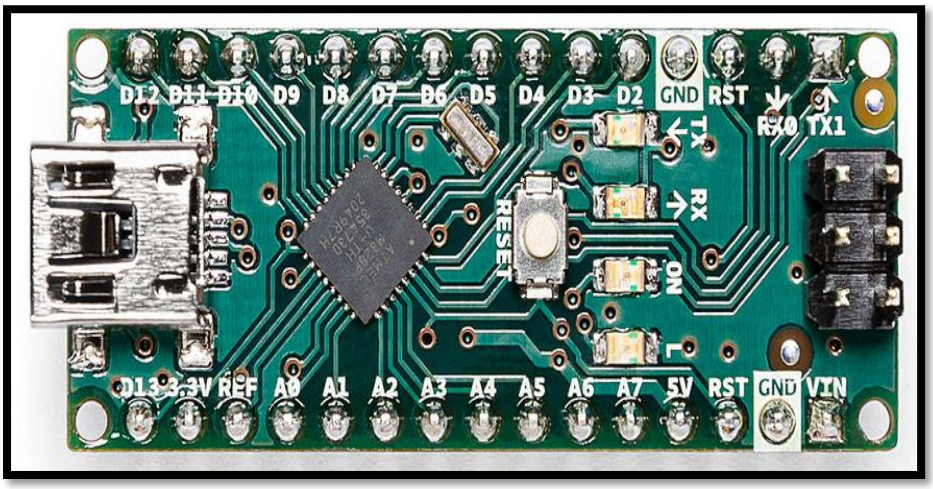

Fig 6.2

Arduino Nano is one type of microcontroller board, and it is designed by Arduino. It can be built with a microcontroller like Atmega328. It is a small-size board and also flexible with a wide variety of applications. This board has many functions and features like an Arduino Duemilanove board. However, this Nano board is different in packaging. It doesn't have any

DC jack so the power supply can be given using a small USB port otherwise straightly connected to the pins like VCC & GND. This board can be supplied with 6 to 20Volts using a mini USB port on the board.

- Flame Sensor

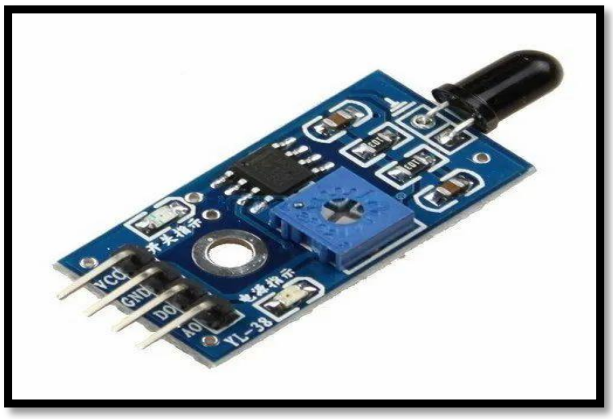

Fig 6.3

A sensor which is most sensitive to a normal light is known as a flame sensor. That's why this sensor module is used in flame alarms. This sensor detects flame otherwise wavelength within the range of 760 nm – 1100 nm from the light source. This sensor can be easily damaged to high temperature. So this sensor can be placed at a certain distance from the flame. The flame detection can be done from a 100cm distance and the detection angle will be 600. The output of this sensor is an analog signal or digital signal. These sensors are used in fire fighting robots like as a flame alarm.

- Relay Module

Fig 6.4

The four-channel relay module contains four 5V relays and the associated switching and isolating components, which makes interfacing with a microcontroller or sensor easy with minimum components and connections. The contacts on each relay are

specified for 250VAC and 30VDC and 10A in each case, as marked on the body of the relays.

Pin Number	Pin Name	Description
1	GND	Ground reference for the module
2	IN1	Input to activate relay 1
3	IN2	Input to activate relay 2
4	IN3	Input to activate relay 3
5	IN4	Input to activate relay 4
6	V_{CC}	Power supply for the relay module
7	V_{CC}	Power supply selection jumper
8	JD-V_{CC}	Alternate power pin for the relay module

- Wireless Relay Module

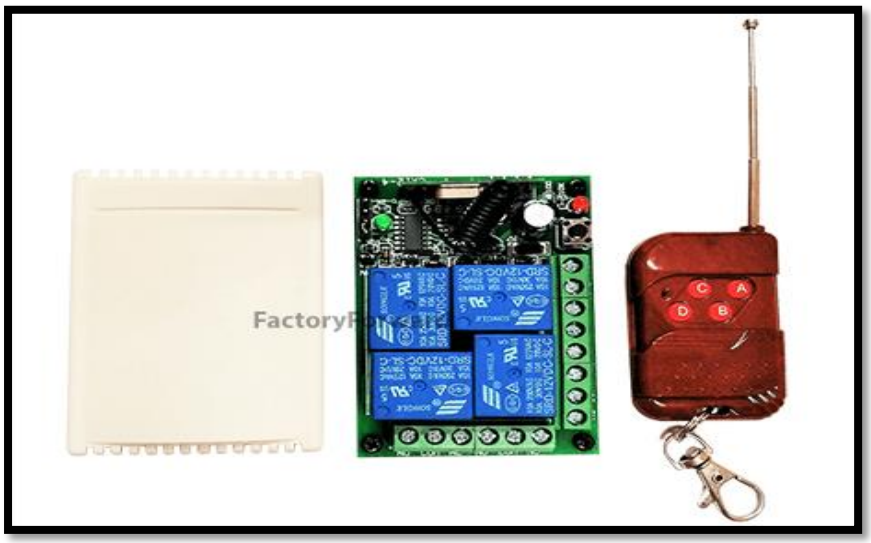

Fig 6.4

433MHz 24V 4 Channel Relay Module Wireless with RF Remote Control Switch without Battery can be controlled by a wireless remote through a wireless RF link. The relays can be used to control 240 V AC up to 10A appliances, motors, other electronic circuits, etc. The board can operate within a range of 50m when the remote is in line of sight and around 25m when indoors. The RF link on this board works on the 433 MHz frequency. Both the transmitter and the receiver come with an on-board antenna and offer a compact and simple design.

- Buzzer

Fig 6.5

An audio signalling device like a beeper or buzzer may be electromechanical or piezoelectric or mechanical type. The main function of this is to convert the signal from audio to sound. Generally, it is powered through DC voltage and used in timers, alarm devices, printers, alarms, computers, etc. Based on the various designs, it can generate different sounds like alarm, music, bell & siren. The pin configuration of the buzzer is shown below. It includes two pins namely positive and negative. The positive terminal of this is represented with the '+' symbol or a longer terminal. This terminal is powered through 6Volts whereas the negative terminal is represented with the '-'symbol or short terminal and it is connected to the GND terminal.

- Resistor

Fig 6.6

A passive electrical component with two terminals that are used for either limiting or regulating the flow of electric current in electrical circuits. The main purpose of resistor is to reduce the current flow and to lower the voltage in any particular portion of the circuit. It is made of copper wires which are coiled around a ceramic rod and the outer part of the resistor is coated with an insulating paint.

- LED’s

Fig 6.7

A light-emitting diode (LED) is a semiconductor device that emits light when an electric current flows through it. When current passes through an LED, the electrons recombine with holes emitting light in the process. LEDs allow the current to flow in the forward direction and blocks the current in the reverse direction. Light-emitting diodes are heavily doped p-n junctions. Based on the semiconductor material used and the amount of doping, an LED will emit coloured light at a particular spectral wavelength when forward biased.

- Connectors

Fig 6.8

Connectors are parts or devices used for electrically connecting or disconnecting circuits etc. They can connect and disconnect by hands or with simple tools without requiring special tools or processes such as soldering. Although there are various types of connectors, the large majority are one-piece types that are used alone or two-piece types consisting of a plug and socket. One-piece types directly connect printed circuit boards and wires to connectors. Two-piece types are divided into plug and socket, the part where the plug and the socket are fitted is called the contact part, and the part where the printed circuit board or the electric wire is attached is called the connection part.

- General Purpose PCB

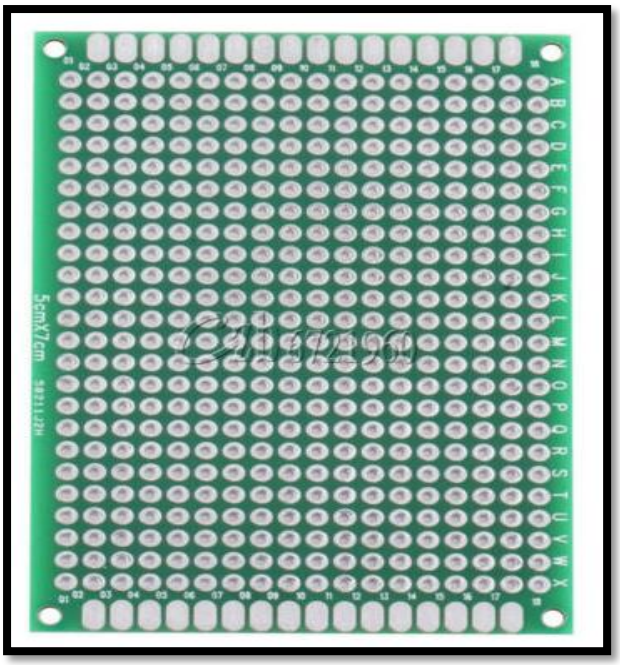

Fig 6.9

General Purpose PCB or Zero PCB or Dotted PCB also known as Perfboard are perfect for quick prototyping and can be used by students for school and college projects. These PCB's are also used by Hobbyist and in Industries for R&D and Testing of small circuits.The PCB provides support as well as electrically connects various Electronic Components in the circuit.

- Bluetooth Module (HC-05)

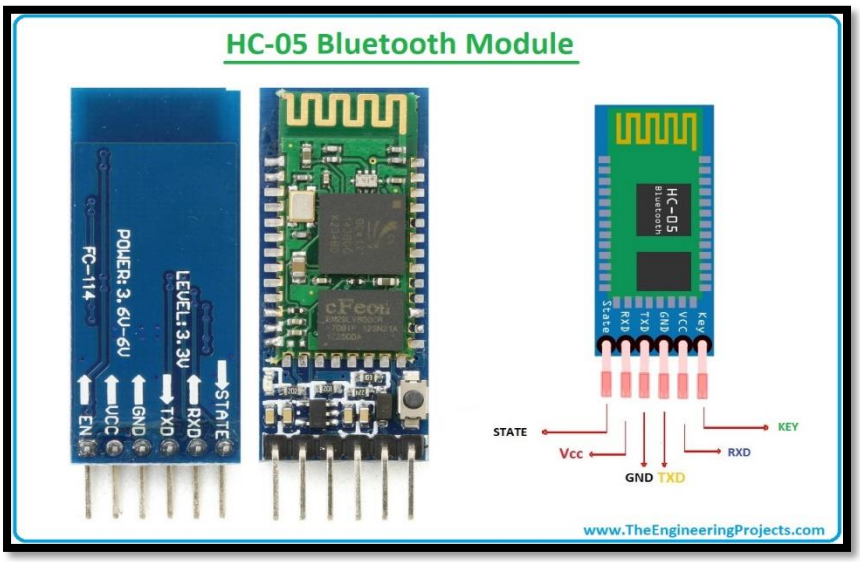

Fig 6.10

HC-05 is a Bluetooth module which is designed for wireless communication. This module can be used in a master or slave configuration. Bluetooth serial modules allow all serial enabled devices to communicate with each other using Bluetooth.

It has 6 pins,

1. Key/EN: It is used to bring Bluetooth module in AT commands mode. If Key/EN pin is set to high, then this module will work in command mode. Otherwise by default it is in data mode. The default baud rate of HC-05 in command mode is 38400bps and 9600 in data mode.

HC-05 module has two modes,

1. Data mode: Exchange of data between devices.

Command mode: It uses AT commands which are used to change setting of HC-05. To send these commands to module serial (USART) port is used.

2. VCC: Connect 5 V or 3.3 V to this Pin.
3. GND: Ground Pin of module.
4. TXD: Transmit Serial data (wirelessly received data by Bluetooth module transmitted out serially on TXD pin)
5. RXD: Receive data serially (received data will be transmitted wirelessly by Bluetooth module).
6. State: It tells whether module is connected or not.

HC-05 has red LED which indicates connection status, whether the Bluetooth is connected or not. Before connecting to HC-05 module this red LED blinks continuously in a periodic manner. When it gets connected to any other Bluetooth device, its blinking slows down to two seconds.

This module works on 3.3V. We can connect 5V supply voltage as well since the module has on board 5 to 3.3 V regulator.

As HC-05 Bluetooth module has 3.3V level for RX/TX and microcontroller can detect 3.3 V level, so, no need to shift transmit level of HC-05 module. But we need to shift the transmit voltage level from microcontroller to RX of HC-05 module.

The data transfer rate of HC-05 module can vary up to 1Mbps is in the range of 10 meters.

- Bike key Unit

Fig 6.11

It is the key unit of the automobile vehicle. It is very vital and important component in this project. It has 4 wire connected to it. The red wire is for supply (positive). The black wire is for ground (negative).the green wire is for earthing.The other is for self-operation of the bike.

- Toggle Switch

Fig 6.12

Toggle Switches have an operating lever that can be pushed up and down or left and right to switch an electrical circuit.

A “toggle” is a small wooden rod that is used as a clothing fastener in the place of buttons. The term “toggle switch” was created because of the resemblance of the switch’s lever to that of the toggle used in clothing.

CHAPTER 7 EXPERIMENTAL OBSERVATIONS

7.1 OBSERVATION:-

LEDS	COLOUR	WORKING	REMOTE KEY SYSTEM
1	GREEN	START VEHICLE	A
2	WHITE	LIGHTS	B
3	RED	BUZZER	C
4	YELLOW	SEAT	D

Table 7.1

CHAPTER 8:- RESULTS AND DISCUSSION

9.1 TROUBLESHOOT

Troubleshooting is a systematic approach to problem-solving that is often used to find and correct issues with project. The first step in most troubleshooting methods is gathering information on the issue, such as an undesired behaviour or a lack of expected functionality.
Once the issue and how to reproduce it are understood, the next step is to eliminate unnecessary components to determine if the issue persists. This can help identify component incompatibility issues and problems caused by third-party products.
Troubleshooting methodologies usually try to isolate a problem so that it can be examined. The initial goal is to diagnose the problem and try obvious solution. Trouble-shooters initially look for common, known causes. Once common issues are ruled out, trouble-shooters must run through a checklist of components to identify where the failure is happening.
The primary objectives of troubleshooting are to figure out why something doesn't work as intended and provide a solution to resolve the issue.
Encountered troubleshoots throughout the project were as follows:

1} Checking for the proper functioning of the Bluetooth app by making a small Bluetooth circuit on breadboard.

The HC-05 module was used here and Arduino uno was interfaced with it. A simple Bluetooth functioning app was developed using the MIT app inverter. After developing the app the HC-05 module was connected with circuitry. The MIT A2 companion was downloaded for the purpose before the actual app was made. The app showed a proper connection to the Bluetooth but the led wouldn’t turn on. This was resolved by interchanging the pins of the transmitter and receiver pins that interfaced with the Bluetooth module.
Firstly a single led was turned on and off next was operate four LEDS. Attempts were made to handle the on-off functioning through a single button and use a separate screen after connecting to the Bluetooth. After many attempts there were many errors introduced like the Bluetooth would be disconnected and not a single led turned on. This was solved by configuring the app into two screens and using separating buttons for on-off and also changing the code within the Arduino code. After all this, all four LEDS were turned on and off simultaneously.
Permission needs to be allowed for both the MIT A2 companion and the app to be downloaded.

2} Simultaneous flame sensor and RFID card code functioning
Both the RFID card and the flame sensor need to be working together, a buzzer was used for the indication of flame sensor. Initially when the code was tested the card was sensed properly but the flame sensor couldn’t work simultaneously with the RFID code. Then separately the flame sensor was tested individually it showed the proper result.
After researching on the code changes were done in the Arduino nano code and so the test procedure were positive.

3}Connectivity of app and relay
When the app was connected to Bluetooth when the main circuit was implemented. then for the off configuration of the buttons, the relays would turn and vice versa.
The problem was solved by changing the code configuration in the mit app inventor.

8.2 FINAL VIEW OF THE PROJECT

1. Main key Unit

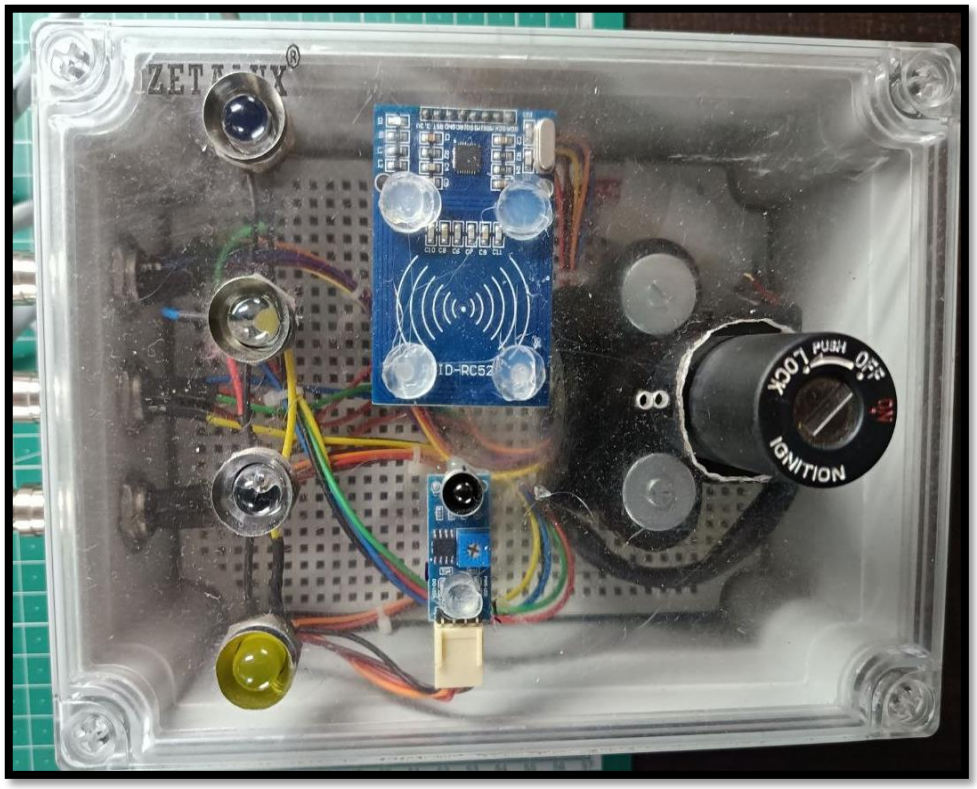

Fig 8.1

2. Control Unit

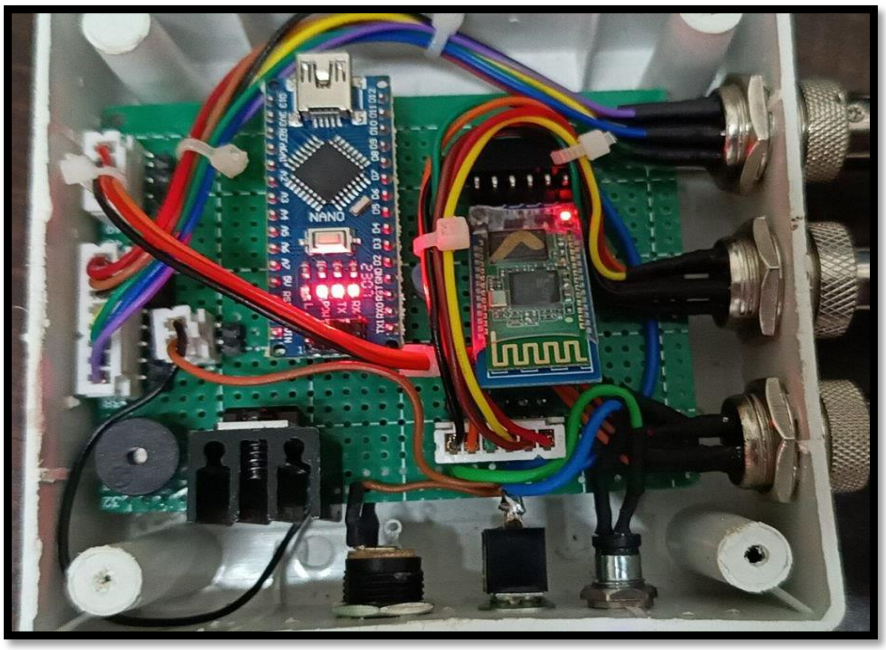

Fig 8.2

3. Relay Unit

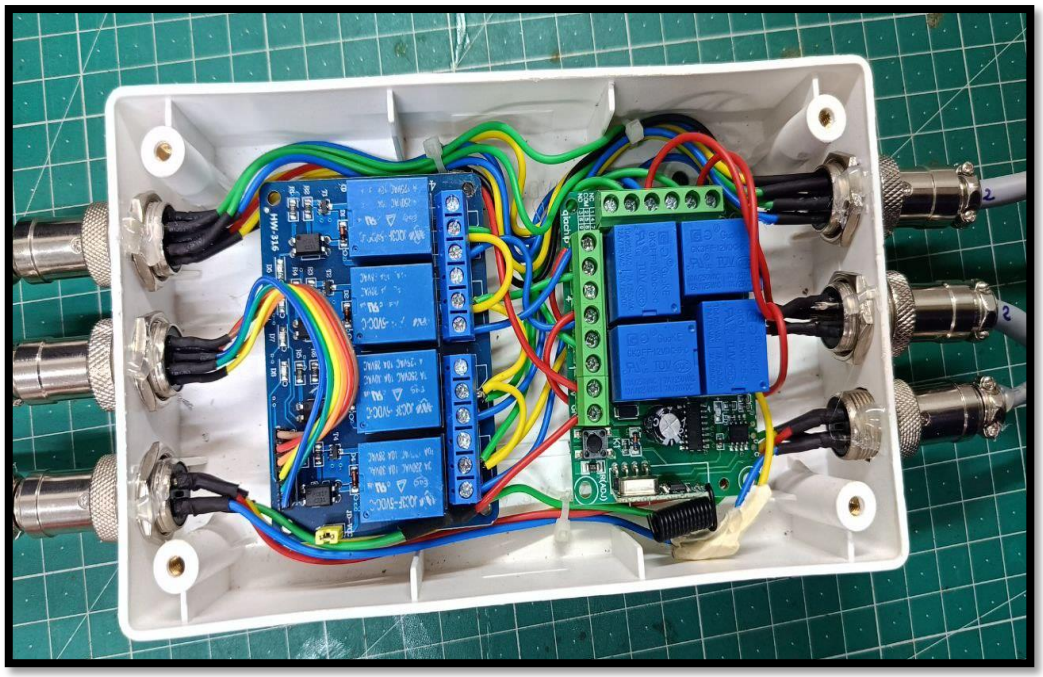

Fig 8.3

CHAPTER 9:- SUMMARY AND CONCLUSIONS

The smart key system for the e-bike has been implemented and tested. The approach provides a stable bidirectional communication with low latency to improve efficiency and portability

The Proposed Work “WIRELESS KEY SYSTEM FOR AUTOMOBILE VEHICLES” Has Been Successfully Tested Which Is The Best Economical, Affordable Energy Solution To Common People. The smart key system for the e-bike has been implemented and tested. The approach provides a stable bidirectional communication with low latency to improve efficiency and portability. All the four configuration were tested successfully.

Since the range of the communication between the key fob and the bike is important, testing was carried out in indoors and outdoors to determine the working range. The bi-directional communication link is established by a button on the key fob. When different buttons are pushed, LED will turn on or off. An app was used to control the state of the bike lights, which is performed without direct human interaction with the bike. In view of requirements, the user and key fob must be able to connect with the bike within a short distance, e.g. house to the garage. The proposed connectivity range is limited to a 20 meters.

If the battery unit of the automobile vehicle is heated due to certain conditions then the buzzer would beep and would instruct the owner .

CHAPTER 10:- FUTURE SCOPE OF WORK

Further integration of the various appliances into communication framework and connected devices have created a special environment- Smart Homes. Even though IoT has been implemented in large systems, several key areas are still yet to take advantage of these communication technologies and remote-control solutions. One of these key areas is the automotive industry. Development of a smart key for an electric bike can potentially provide consumer with the ability to monitor the status of the bike and control effectively the main features.

The concept of a smart key that is designed to enrich the usability of an electric bike, to inform the owner about the vehicle status, its current operating conditions and control important security functions can also be done. Further integration of the various appliances into communication framework and connected devices have created a special environment- Smart Homes.Even though IoT has been implemented in large systems, several key areas are still yet to take advantage of these communication technologies and remote-control solutions.

Through the development of technologies, the traditional pin tumbler lock is slowly being transitioned to a remotely controlled electronic key in automotive systems. A remotely controllable key with in-built functions, such as battery lifecycle monitor, light control and live updates from the security system can be monitored remotely without the need of checking the vehicle in person. key has been developed to remotely controllable key.

REFERNCE

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https://www.google.com/search?q=connectors+female+to+female&tbm=isch&ved=2ahUKEwiQxOWV09n-AhU36HMBHQ47AlAQ2-cCegQIABAA&oq=connectors+female+to+female&gs_lcp=CgNpbWcQAzoECCMQJzoHCAAQigUQQzoICAAQgAQQsQM6BQgAEIAEOgYIABAFEB46BggAEAgQHlCuBViXP2C8QWgBcAB4AIABmQOIAd0gkgEJMC44LjkuMS4xmAEAoAEBqgELZ3dzLXdpei1pbWfAAQE&sclient=img&ei=nZZSZNCBMbfQz7sPjvaIgAU&rlz=1C1CHBD_enIN886IN886#imgrc=n6cexc2RUhEf4